

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

WINTER SEMESTER, 2023-2024
FULL MARKS: 75

CSE 4309: Theory of Computing

Programmable calculators are not allowed. Do not write anything on the question paper.
 Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) State the differences between a Deterministic Finite Automaton (DFA) and Nondeterministic Finite Automaton (NFA). Explain δ for DFA, NFA, and epsilon NFA (ϵ -NFA). 5
(CO1)
(PO1)
- b) Design a DFA for accepting a set of strings over input alphabet $\Sigma = \{a, b\}^*$ for the language $L = \{a^2 b w a^2 \mid w \in \{a, b\}^*\}$. 8
(CO2)
(PO1)
 - c) Finite automata are good models for computer with an extremely limited amount of memory. The controller for an automatic door is one such example. Show the state diagram of this finite automata with the required states and input conditions. 6
(CO2)
(PO1)
 - d) The classic game Pac-Man requires the player to navigate through a maze, eating pellets, and avoiding the ghosts who chase him through the maze. Occasionally, Pac-Man can turn the tables on his pursuers by eating a power pellet, which temporarily grants him the power to eat the ghosts. When this occurs, the ghosts behavior changes and instead of chasing Pac-Man, they try to avoid him. The ghosts in Pac-Man have four behaviors: randomly wander the maze, chase Pac-Man when he is within line of sight, flee Pac-Man after Pac-Man has consumed a power pellet, and return to the central base to regenerate. Draw the state diagram of a finite automata that emulates the behavior of the ghosts in Pac-Man. Show the states that the ghosts might be in at any given moment and also what inputs they take to make a transition from one state to another state. 6
(CO2)
(PO2)
2. a) What are the applications of regular expressions and finite automata? 5
(CO1)
(PO1)
- b) Assume the password policy for creating your student account is set as follows:
 - Password should start with atleast one symbols such as !, #, \$, @, etc.
 - Password should contain atleast one upper case letter or atleast one digit as substring.
 - Password can contain zero or more number of lower case letters.
 Derive the regular expression that generates the passwords based on the policies mentioned above and design a Finite Automata that accepts the strings generated by the regular expression. 6
(CO2)
(PO2)
 - c) You are given an NFA, $N_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ that accepts the language A, and an NFA, $N_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ that accepts the language B. Show that there exists an NFA, N that recognizes the language $A \circ B$. 6
(CO1)
(PO1)
 - d) Convert the following Regular Expressions to equivalent NFAs: 4 + 4
(CO5)
(PO1)
 - i. $((00)^*(11) \cup 01)^*$
 - ii. $(0 \cup 10)^* 010 (0 \cup 1)^*$

3. a) Show that language $L = \{ww \mid w \in \{0, 1\}^*\}$ is not regular using the Pumping lemma of regular languages. 8
(CO1)
(PO1)
- b) Minimize the DFA shown in Figure 1 using the Equivalence theorem. 7
(CO3)
(PO2)

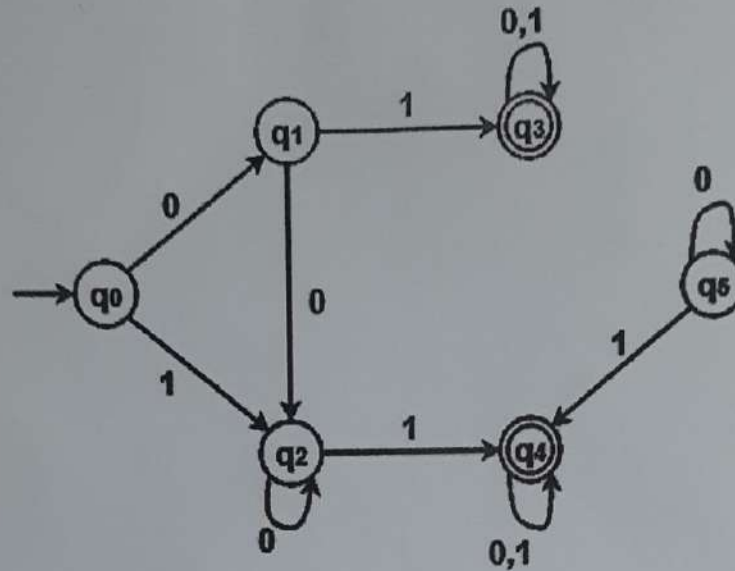


Figure 1: DFA State Diagram for Question 3.b

- c) Convert the ϵ -NFA shown in Table 1 to an equivalent NFA by computing the ϵ -closure of each state. 10
(CO5)
(PO1)

Table 1: Transition Table for Question 3.c

	a	b	ϵ
$\rightarrow q_0$	$\emptyset$	$\{q_0\}$	$\{q_1\}$
q_1	$\{q_1\}$	$\emptyset$	$\{q_2\}$
$* q_2$	$\{q_1\}$	$\{q_0\}$	$\emptyset$